

Supplementary Information

Machine Learning Applied to Aquatic Toxicity and Chemical Reactions of Xenobiotic Compounds

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Selected (standard) OH radical attack reactions, assembled in MarvinSketch software and their respective SMARTS codes

Alkane attack reactions

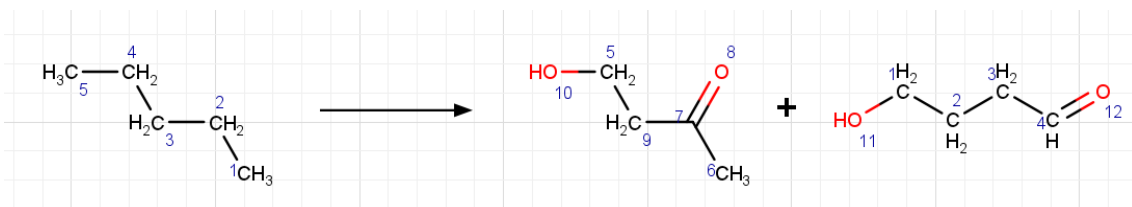


Figure S1. Representation of the OH radical attack reaction on alkanes (SMARTS 1 reaction).

SMARTS code: [*6:1]-[*6:2]-[*6:3]-[*6:4]-[*6:5]>>[*6:6]-[*6:7](=[O:8])-[*6:9]-[*6:5]-[*8:10].[*8:11]-[*6:1]-[*6:2]-[*6:3]-[*6:4]=[O:12]

Aromatic attack reactions

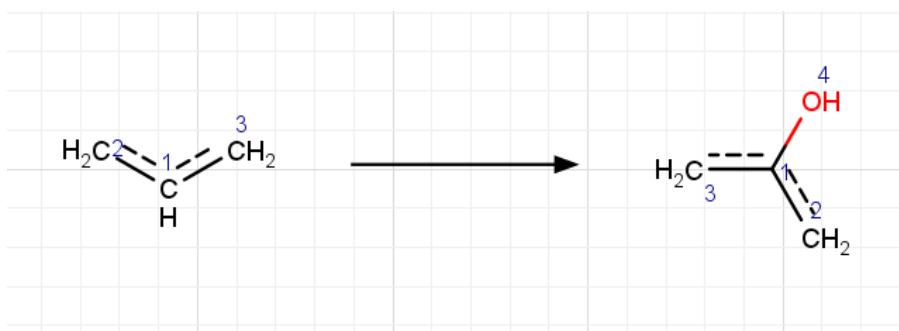


Figure S2. Representation of the OH radical attack reaction on aromatic compounds, with the formation of phenols (SMARTS 2 reaction).

SMARTS code: [c:2]:[c:1]:[c:3]>>[*8:4]-[c:1](:[c:2]):[c:3]

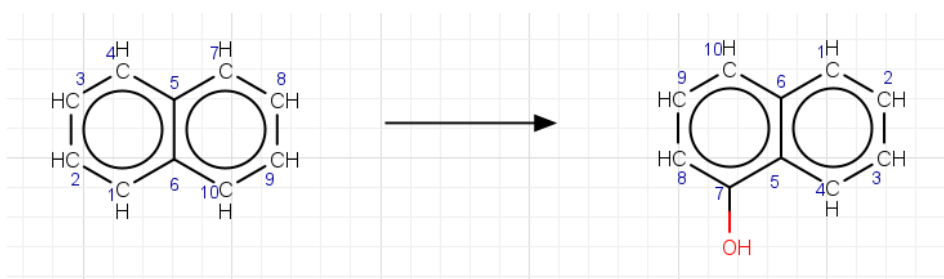


Figure S3. Representation of the OH radical attack reaction on naphthenic aromatic compounds (SMARTS 3 reaction).

SMARTS code: [c:2]1[c:3][c:4][c:5]2[c:7][c:8][c:9][c:10][c:6]2[c:1]1>>[*8]-[c:7]1[c:8][c:9][c:10][c:6]2[c:1][c:2][c:3][c:4][c:5]12

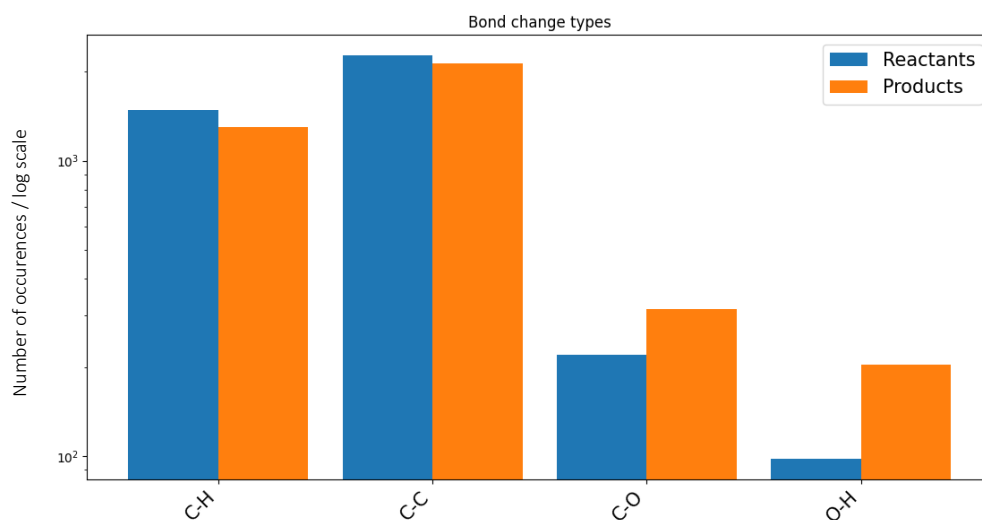


Figure S4. Bond change types. The number of times each type of bond switching occurs in SMARTS 3 reaction.

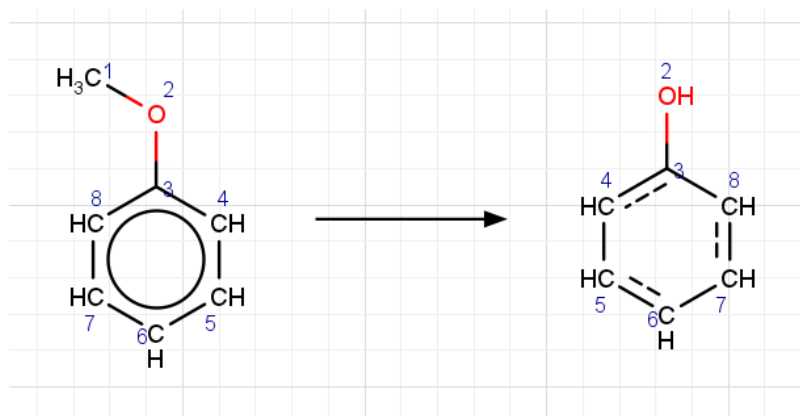


Figure S5. Representation of the OH radical attack reaction on aromatic compounds with the formation of phenols (SMARTS 4 reaction).

SMARTS code: [#6]-[#8:2]-[c:3]1[c:4][c:5][c:6][c:7][c:8]1>>[#8:2]-[c:3]1[c:4][c:5][c:6][c:7][c:8]1

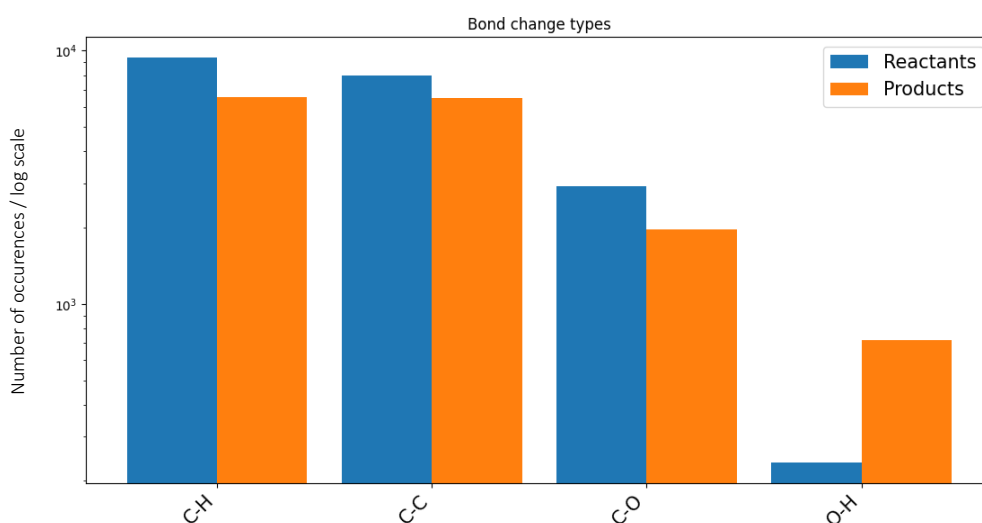


Figure S6. Bond change types. The number of times each type of bond switching occurs in SMARTS 4 reaction.

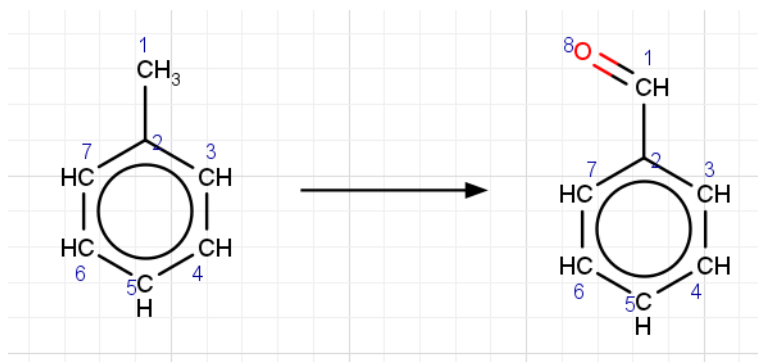


Figure S7. Representation of the OH radical attack reaction on aromatic compounds (SMARTS 5 reaction).

SMARTS code: [#6:1]-[c:2]1[c:3][c:4][c:5][c:6][c:7]1>>[O:8]=[#6:1]-[c:2]1[c:3][c:4][c:5][c:6][c:7]1

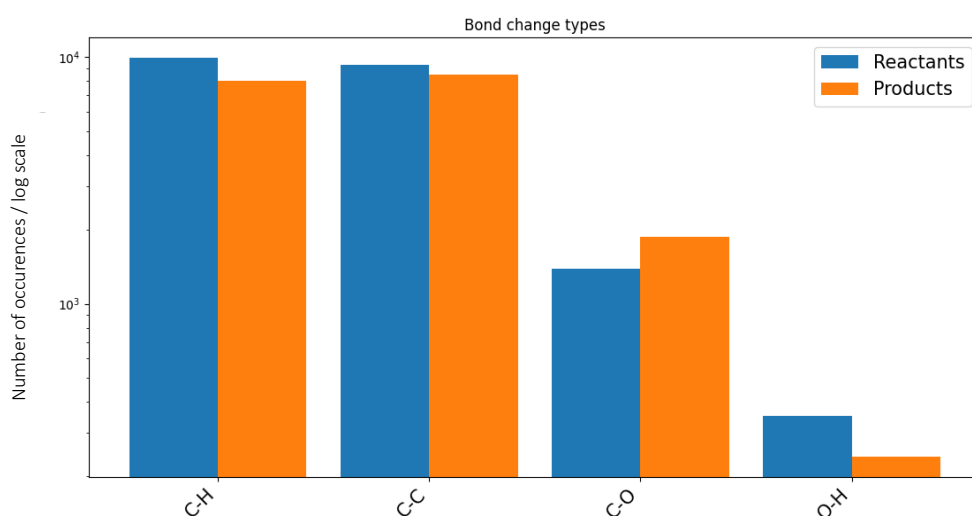


Figure S8. Bond change types. The number of times each type of bond switching occurs in SMARTS 5 reaction.

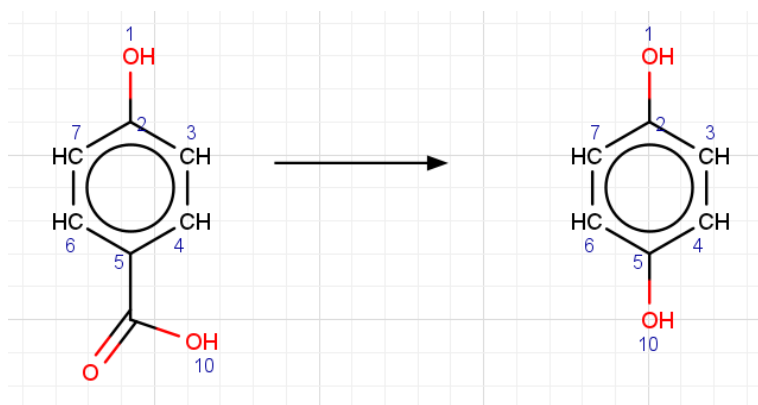


Figure S9. Representation of the OH radical attack reaction on aromatic compounds with the formation of diphenols (SMARTS 6 reaction).

SMARTS code: [#8:10]-[#6](=O)-[c:5]1[c:4][c:3][c:2](-[#8:1])[c:7][c:6]1>>[#8:1]-[c:2]1[c:3][c:4][c:5](-[#8:10])[c:6][c:7]1

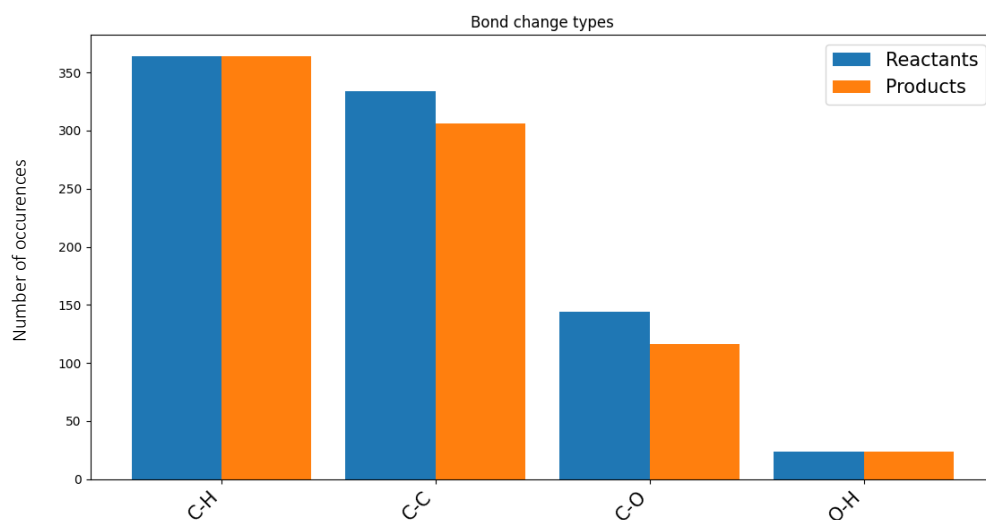


Figure S10. Bond change types. The number of times each type of bond switching occurs in SMARTS 6 reaction.

Attack on aliphatic alkenes

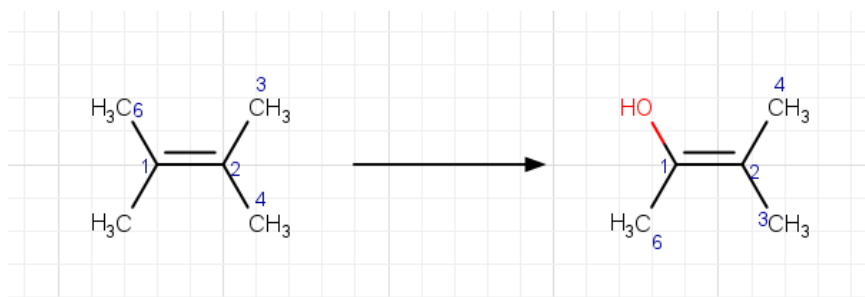


Figure S11. Representation of the OH radical attack reaction on aliphatic alkenes (SMARTS 7 reaction).

SMARTS code: [#6:3]\[#6:2](-[#6:4])=[#6:1](\[#6])-[#6:6]>>[#6:3]\[#6:2](-[#6:4])=[#6:1](/[#6:6])-[#8]

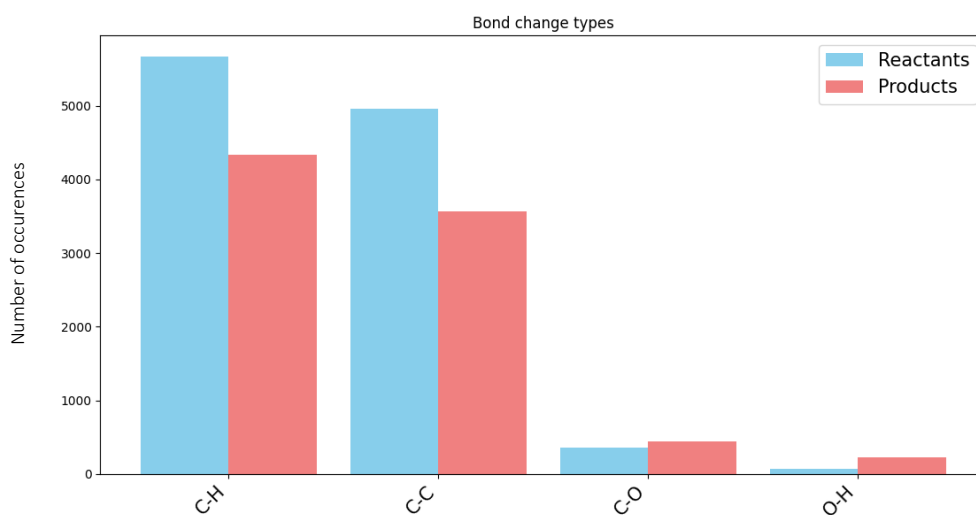


Figure S12. Bond change types. The number of times each type of bond switching occurs in SMARTS 7 reaction.

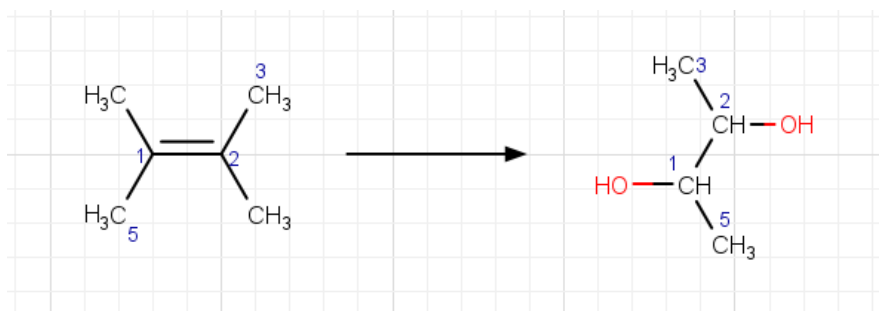


Figure S13. Representation of the OH radical attack reaction on aliphatic alkenes (SMARTS 8 reaction).

SMARTS code: [#6:3][#6:2](-[#6])=[#6:1](\[#6:5])-[#6]>>[#6:3]-[#6:2](-[#8])-[#6:1](-[#6:5])-[#8]

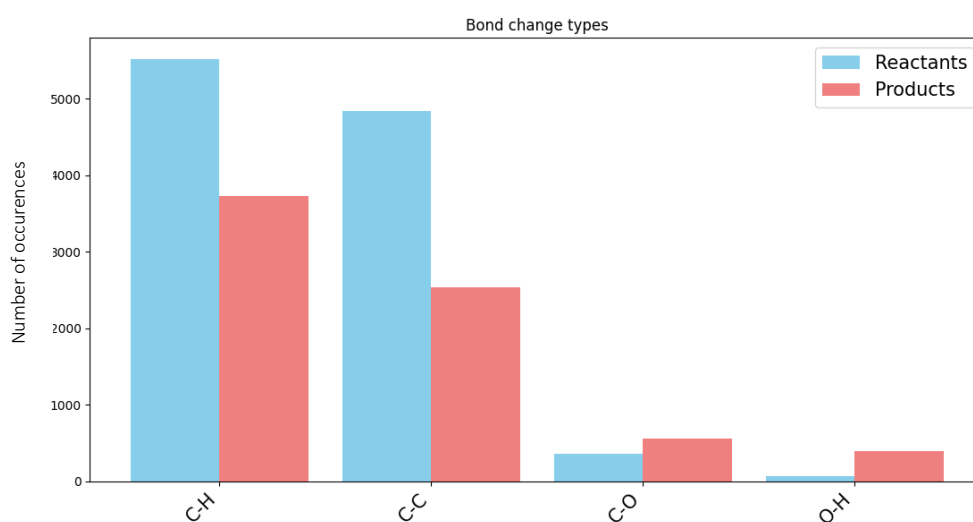


Figure S14. Bond change types. The number of times each type of bond switching occurs in SMARTS 8 reaction.

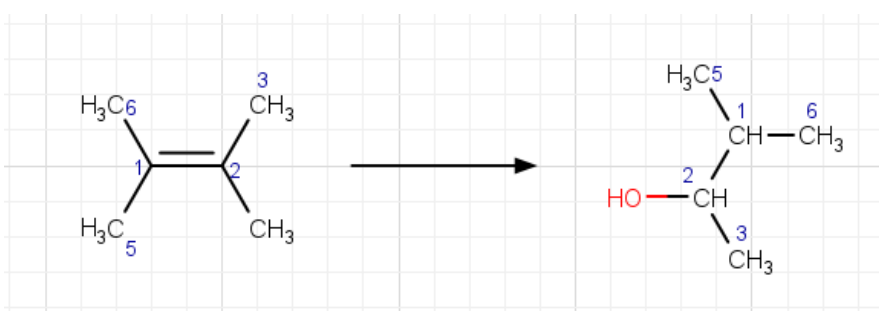


Figure S15. Representation of the OH radical attack reaction on aliphatic alkenes (SMARTS 9 reaction).

SMARTS code: [#6:3][#6:2](-[#6])=[#6:1](\[#6:5])-[#6:6]>>[#6:5]-[#6:1](-[#6:6])-[#6:2](-[#6:3])-[#8]

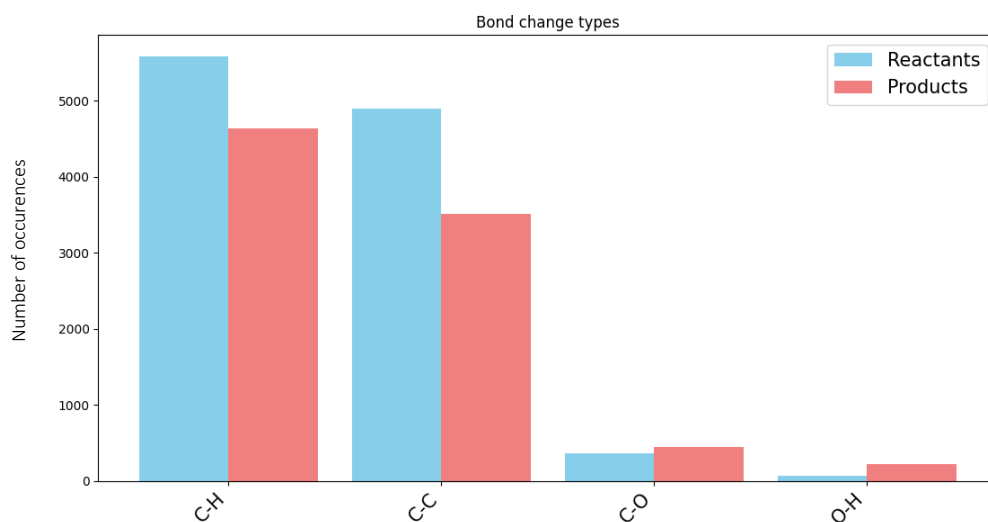


Figure S16. Bond change types. The number of times each type of bond switching occurs in SMARTS 9 reaction.

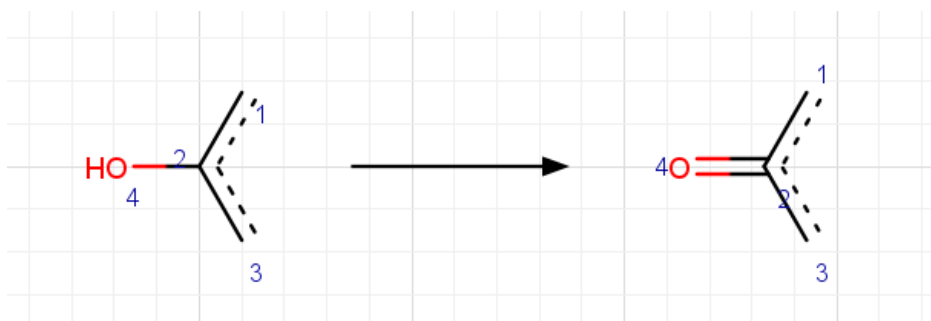


Figure S17. Representation of the OH radical attack reaction on aliphatic alkenes (SMARTS 10 reaction).

SMARTS code:[#6:1][#6:2](=,[#6:3])-[#8:4]>>[#6:1][#6:2]([#6:3])=[O:4]

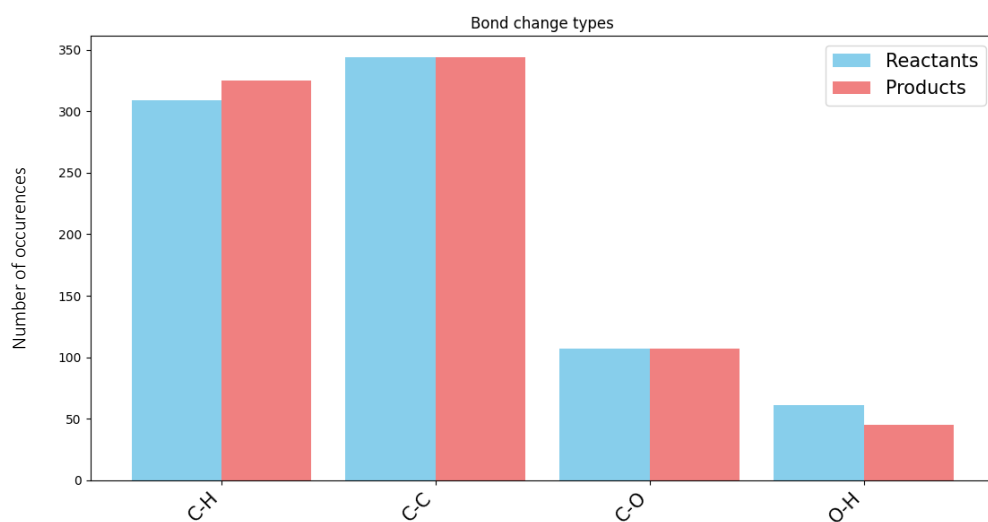


Figure S18. Bond change types. The number of times each type of bond switching occurs in SMARTS 10 reaction.

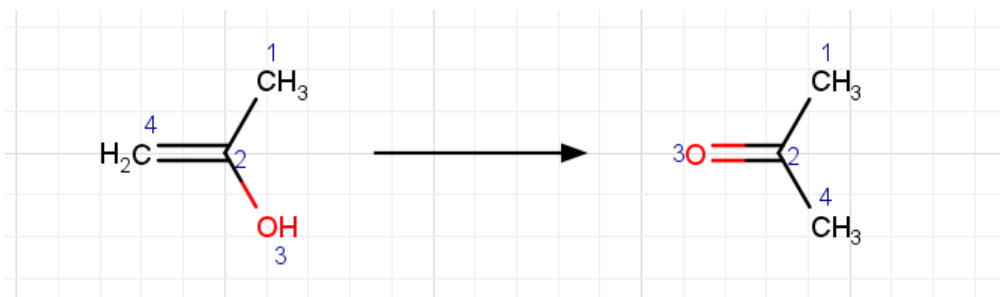


Figure S19. Representation of the OH radical attack reaction on aliphatic alkenes (SMARTS 11 reaction).

SMARTS code: [#6:1]-[#6:2](-[#8:3])=[#6:4]>>[#6:1]-[#6:2](-[#6:4])=[O:3]

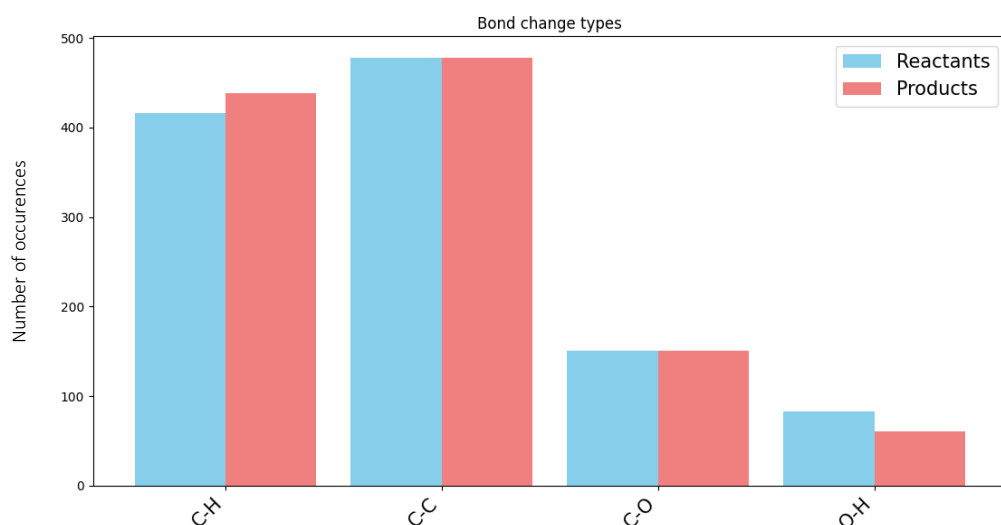


Figure S20. Types of bond switching. The number of times each type of bond switching occurs in SMARTS 11 reaction.

Attack on nitrogenous groups

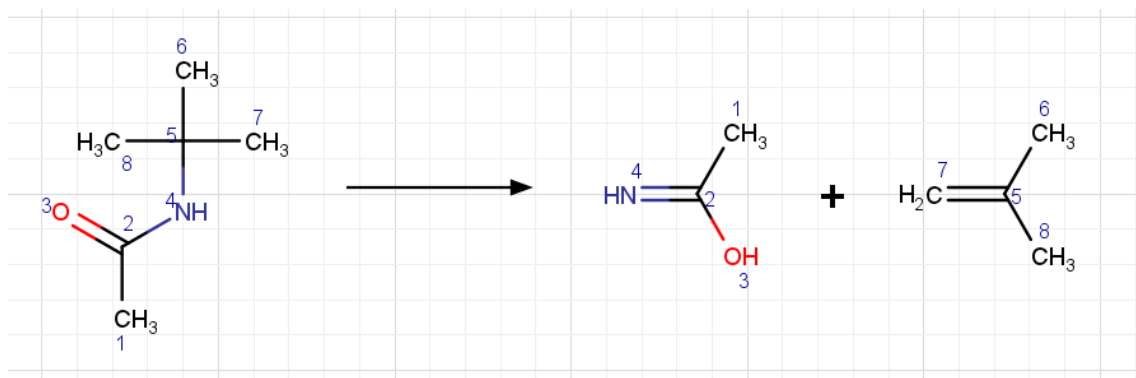


Figure S21. Representation of the attack reaction of the OH radical on nitrogen groups (SMARTS 12 reaction).

SMARTS code: [#6:1]-[#6:2](=[O:3])-[#7:4][C:5]([#6:6])([#6:7])[#6:8]>>[#6:1]-[#6:2](-[#8:3])=[#7:4].[#6:6]-[#6:5](-[#6:8])=[#6:7]

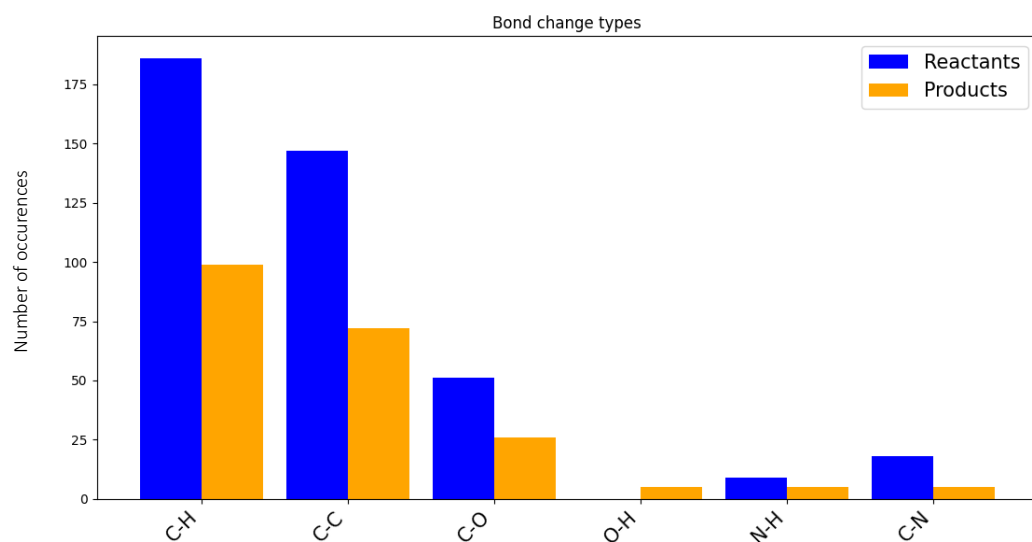


Figure S22. Bond change types. The number of times each type of bond switching occurs in SMARTS 12 reaction.

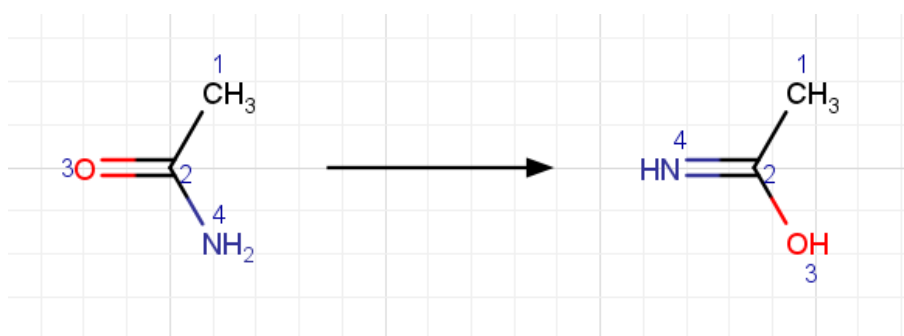


Figure S23. Representation of the OH radical attack reaction on nitrogen groups (SMARTS 13 reaction).

SMARTS code: [#6:1]-[#6:2](-[#7:4])=[O:3]>>[#6:1]-[#6:2](-[#8:3])=[#7:4]

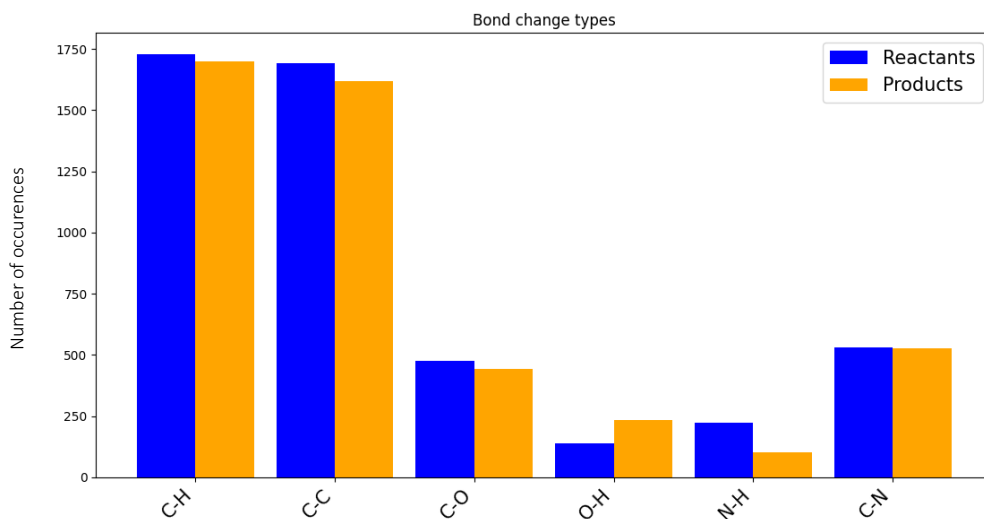


Figure S24. Bond change types. The number of times each type of bond switching occurs in SMARTS 13 reaction.

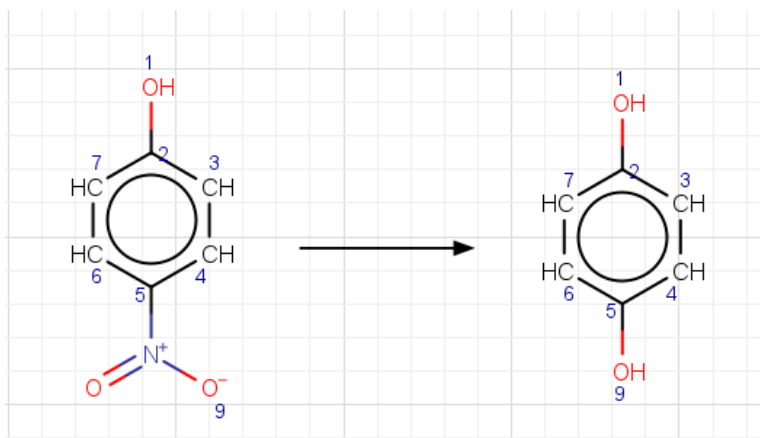


Figure S25. Representation of the attack reaction of the OH radical on nitrogen groups (SMARTS 14 reaction).

SMARTS code: [#8:1]-[c:2]1[c:3][c:4][c:5]([c:6][c:7]1)-[#7+]-[#8-:9]=O>>[#8:1]-[c:2]1[c:3][c:4][c:5](-[#8:9])([c:6][c:7]1

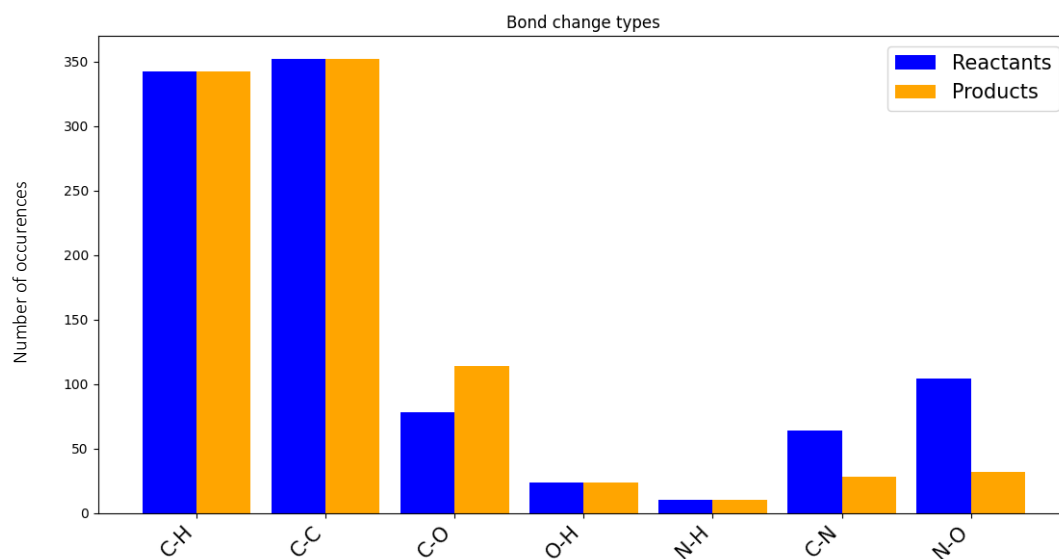


Figure S26. Bond change types. The number of times each type of bond switching occurs in SMARTS 14 reaction.

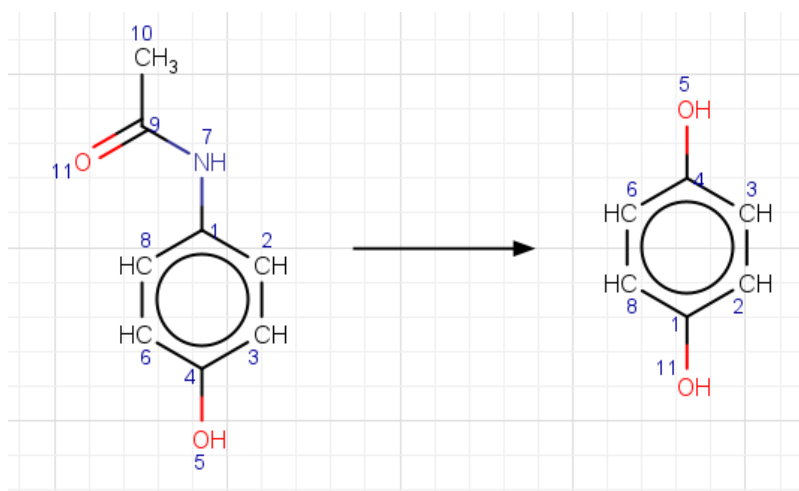


Figure S27. Representation of the attack reaction of the OH radical on nitrogen groups (SMARTS 15 reaction).

SMARTS code: [#6:10]-[#6:9](=[O:11])-[#7:7]-[c:1]1[c:2][c:3][c:4](-[#8:5])[c:6][c:8]1>>[#8:5]-[c:4]1[c:3][c:2][c:1](-[#8:11])[c:8][c:6]1

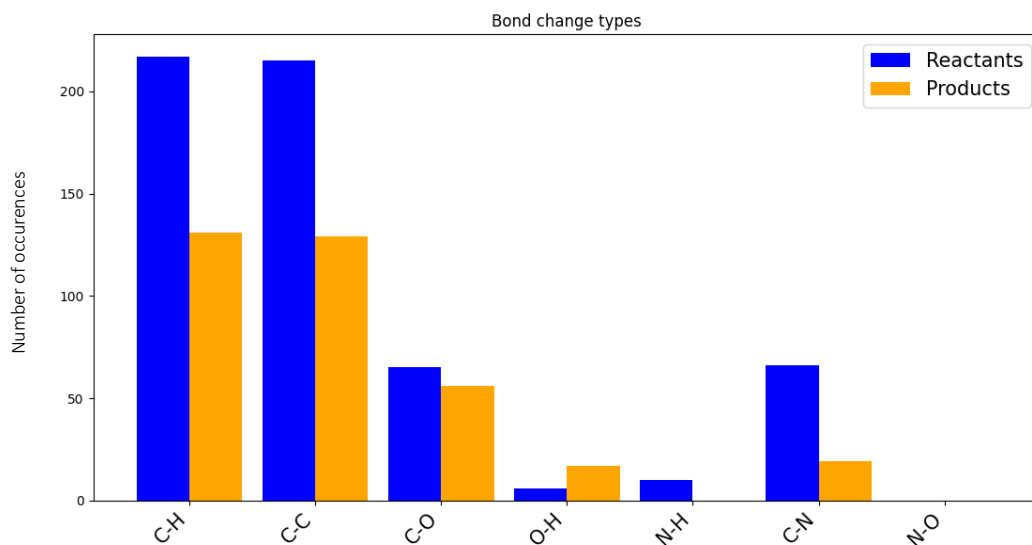


Figure S28. Types of bond switching. The number of times each type of bond switching occurs in SMARTS 15 reaction.

Attack on organochlorines

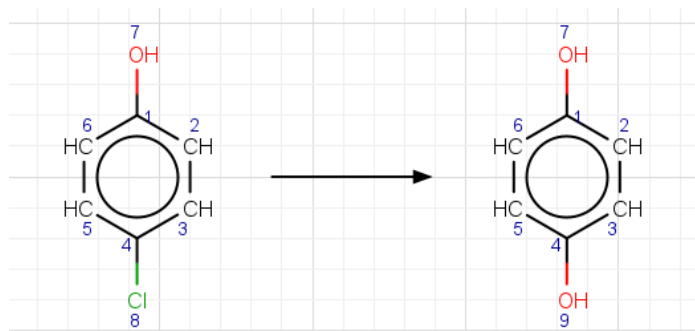


Figure S29. Representation of the OH radical attack reaction on organochlorines (SMARTS 16 reaction).

SMARTS code: [#8:7]-[c:1]1[c:2][c:3][c:4]([Cl:8])[c:5][c:6]1>>[#8:7]-[c:1]1[c:2][c:3][c:4](-[#8:9])[c:5][c:6]1

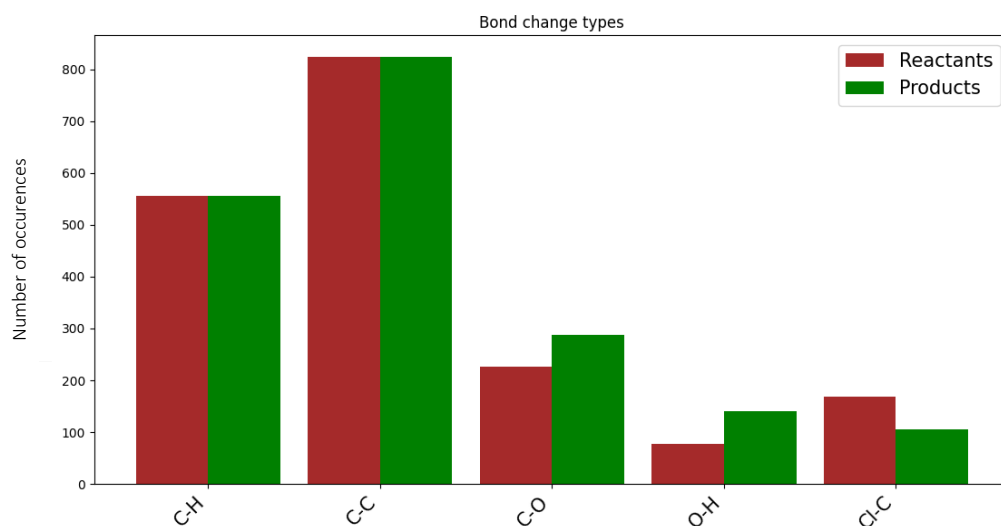


Figure S30. Bond change types. The number of times each type of bond switching occurs in SMARTS 16 reaction.

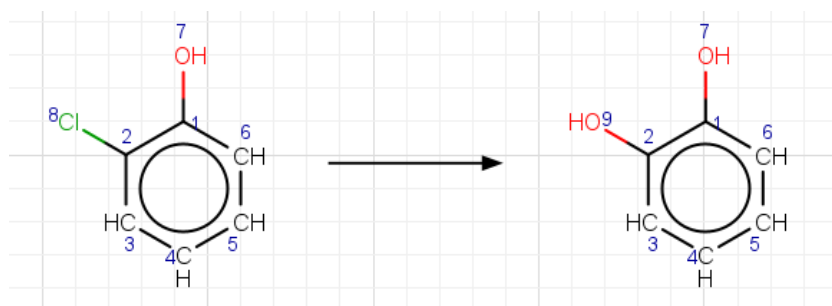


Figure S31. Representation of the OH radical attack reaction on organochlorines (SMARTS 17 reaction).

SMARTS code: [#8:7]-[c:1]1[c:6][c:5][c:4][c:3][c:2]1[Cl:8]>>[#8:7]-[c:1]1[c:6][c:5][c:4][c:3][c:2]1-[#8:9]

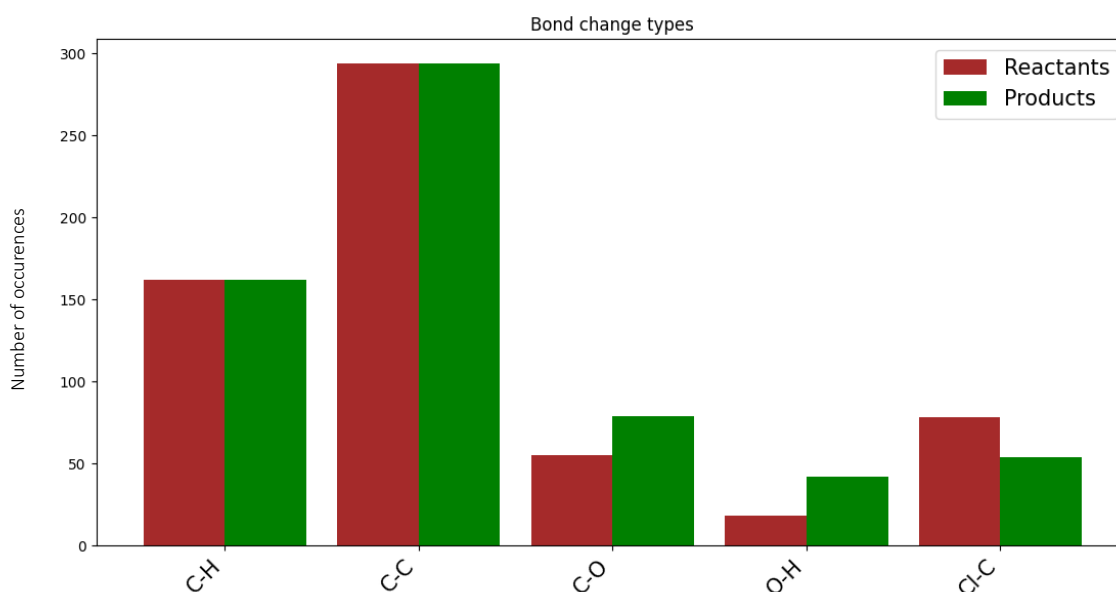


Figure S32. Types of bond switching. The number of times each type of bond switching occurs in SMARTS 17 reaction.

Table S1. Evaluation of QSAR variables using variable selection resources/methods for the 44 most important variables used in the Extra Tree Regressor model

Feature/descriptor	ID	Importance	Pearson coefficient	Mutual information	RFECV
MolLogP	1	0.0762	0.5345	0.2635	true
BCUT2D_LOGPHI	2	0.0190	0.4344	0.2118	true
PEOE_VSA6	3	0.0164	0.4454	0.1815	true
fr_COO	4	0.0161	-0.2126	0.0465	true
FractionCSP3	5	0.0156	-0.1316	0.0611	true
fr_COO2	6	0.0143	-0.2126	0.0284	true
SlogP_VSA12	7	0.0101	0.4319	0.1484	true
EState_VSA9	8	0.0080	0.4878	0.1350	true
fr_allylic_oxid	9	0.0063	0.2269	0.0386	true
BCUT2D_CHGHI	10	0.0061	0.3229	0.0787	true
SlogP_VSA4	11	0.0060	0.2118	0.0663	true
SMR_VSA7	12	0.0053	0.2954	0.0703	true
fr_halogen	13	0.0050	0.3957	0.1061	true
MinAbsEStateIndex	14	0.0045	-0.2050	0.0479	true
NOCcount	15	0.0043	-0.0995	0.0265	true
fr_nitro_arom	16	0.0038	0.0862	0.0304	true
Chi2v	17	0.0038	0.4381	0.2390	true
fr_bicyclic	18	0.0035	0.2488	0.0261	true
BCUT2D_LOGPLOW	19	0.0032	-0.0908	0.0670	true
SMR_VSA9	20	0.0031	0.1664	0.0597	true
Chi4v	21	0.0029	0.4403	0.2381	true
Phi	22	0.0028	0.3603	0.0971	true
fr_aldehyde	23	0.0027	-0.0305	0.0014	true
Chi0v	24	0.0026	0.4219	0.2612	true
HallKierAlpha	25	0.0026	0.0324	0.1134	true
PEOE_VSA2	26	0.0025	-0.0402	0.00	true
NumAromaticRings	27	0.0025	0.1923	0.0849	true
fr_aryl_methyl	28	0.0023	-0.0197	0.0292	true
HeavyAtomMolWt	29	0.0023	0.4076	0.1758	true
fr_ester	30	0.0022	0.1932	0.0173	true
EState_VSA4	31	0.0020	0.1818	0.0213	true
BCUT2D_CHGLO	32	0.0020	-0.2279	0.1533	true
Kappa3	33	0.0019	-0.0208	0.0075	true
BCUT2D_MRLOW	34	0.0017	-0.0323	0.0348	true
SlogP_VSA3	35	0.0017	0.1038	0.0118	true
MaxPartialCharge	36	0.0016	0.0587	0.1072	true
EState_VSA10	37	0.0016	-0.0036	0.00	true

VSA_EState10	38	0.0016	0.4093	0.1576	true
Kappa1	39	0.0014	0.3663	0.1843	true
fr_Ar_OH	40	0.0014	0.0083	0.0091	true
EState_VSA8	41	0.0013	-0.0396	0.0529	true
SlogP_VSA11	42	0.0013	0.1435	0.0693	true
fr_phenol_noOrthoHbond	43	0.0011	0.0225	0.0011	true
MinAbsPartialCharge	44	0.0010	0.0448	0.0632	true

RFECV: recursive feature elimination with cross-validation; MolLogP: molecular octanol/water partition coefficient (logP); BCUT2D_LOGPHI: BCUT two-dimensional molecular descriptor weighted by LOGPHI; PEOE_VSA6: partial charge-weighted van der Waals surface area descriptor; fr_COO: carboxylic acid or carboxylate group; fractionCSP3: fraction of sp³-hybridized carbon atoms; fr_COO2: ester or carboxylic acid derivative; SlogP_VSA12: van der Waals surface area associated with atomic LogP contribution range 12; EState_VSA9: van der Waals surface area associated with electrotopological state range 9; fr_allylic_oxid: allylic oxidation site; BCUT2D_CHGHI: highest eigenvalue of the Burden matrix weighted by atomic partial charges; SlogP_VSA4: van der Waals surface area associated with atomic LogP contribution range 4; SMR_VSA7: van der Waals surface area associated with molar refractivity range 7; fr_halogen: halogen-containing fragment; MinAbsEStateIndex: minimum absolute electrotopological state index; NOCount: number of nitrogen and oxygen atoms; fr_nitro_arom: aromatic nitro group; Chi2v: valence connectivity index of order 2; fr_bicyclic: bicyclic system; BCUT2D_LOGPLOW: lowest eigenvalue of the Burden matrix weighted by atomic LogP contributions; SMR_VSA9: van der Waals surface area associated with molar refractivity range 9; Chi4v: valence connectivity index of order 4; Phi: Kier molecular flexibility index; fr_aldehyde: aldehyde group; Chi0v: valence connectivity index of order 0; HallKierAlpha: Hall-Kier alpha index; PEOE_VSA2: van der Waals surface area associated with PEOE partial charge range 2; NumAromaticRings: number of aromatic rings; fr_aryl_methyl: aryl methyl group; HeavyAtomMolWt: molecular weight considering only heavy atoms; fr_ester: ester group; EState_VSA4: van der Waals surface area associated with electrotopological state range 4; BCUT2D_CHGLO: lowest eigenvalue of the Burden matrix weighted by atomic partial charges; Kappa3: third Kier molecular shape index; BCUT2D_MRLOW: lowest eigenvalue of the Burden matrix weighted by atomic molar refractivity contributions; SlogP_VSA3: van der Waals surface area associated with atomic LogP contribution range 3; MaxPartialCharge: maximum atomic partial charge; EState_VSA10: van der Waals surface area associated with electrotopological state range 10; VSA_EState10: van der Waals surface area weighted by electrotopological state values; Kappa1: first Kier molecular shape index; fr_Ar_OH: aromatic hydroxyl group; EState_VSA8: van der Waals surface area associated with electrotopological state range 8; SlogP_VSA11: van der Waals surface area associated with atomic LogP contribution range 11; fr_phenol_noOrthoHbond: phenol lacking intramolecular ortho hydrogen bonding; MinAbsPartialCharge: minimum absolute atomic partial charge.

Table S2. Evaluation metrics for ET regression model with Fingerprints as input

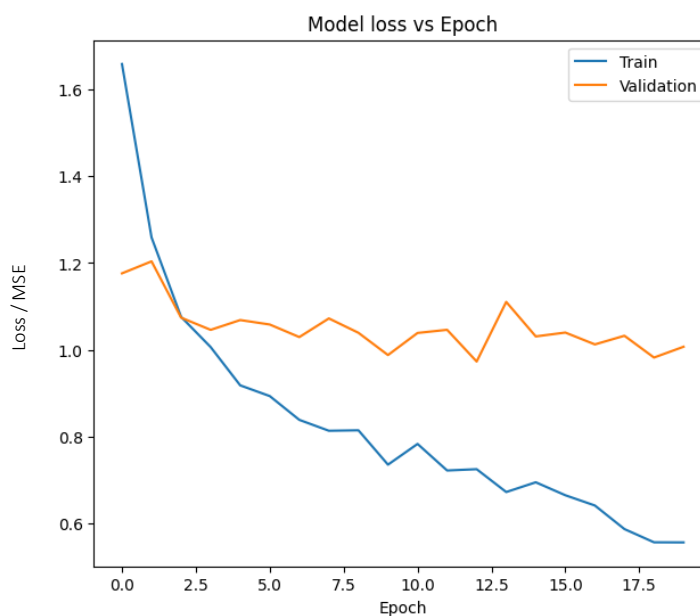
Fingerprints	Sets	MSE	RMSE	MAE	R ² score
MACCS	calibration	0.0984	0.3137	0.1948	0.9527
	validation	1.8451	1.3583	1.0494	0.2660
Morgan, size = 512	calibration	0.2272	0.4766	0.3444	0.8908
	validation	2.0178	1.4205	1.1296	0.1973
Morgan, size = 1024	calibration	0.2273	0.4767	0.3444	0.8908
	validation	1.9853	1.4090	1.1230	0.2103
Morgan, size = 2048	calibration	0.2265	0.4759	0.3457	0.8912
	validation	2.0105	1.4179	1.1293	0.2002
RDKit, size = 1024	calibration	0.0989	0.3145	0.1825	0.9525
	validation	1.9572	1.3989	1.1057	0.2215

MSE: mean squared error; RMSE: root mean square error; MAE: mean absolute error; R² score: coefficient of determination; MACCS: Molecular ACCess System.

Table S3. Evaluation metrics for the Keras NN model

Sets	MSE	RMSE	MAE	R ² score
calibration	0.5178	0.7196	0.5361	0.7518
validation	1.129	1.063	0.7749	0.5401

MSE: mean squared error; RMSE: root mean square error; MAE: mean absolute error; R² score: coefficient of determination.

**Figure S33.** Loss function in relation to the number of epochs considering the early stopping feature.

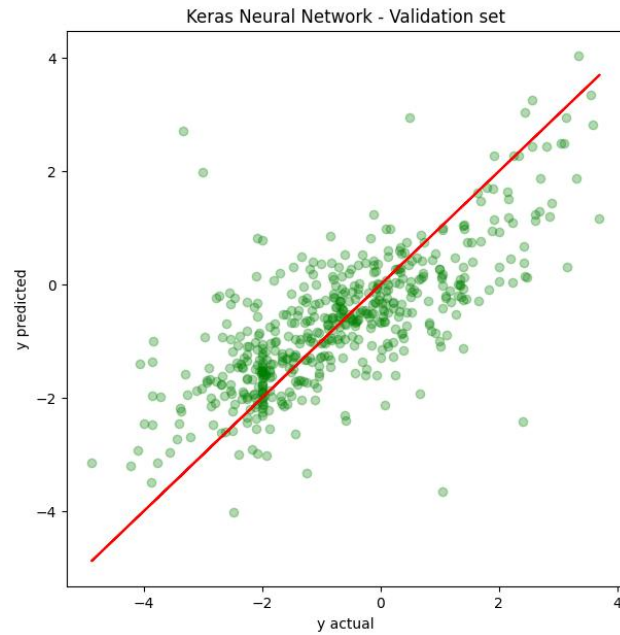


Figure S34. Graphical visualization of the results predicted by the KNN model *versus* the actual results for the validation set.

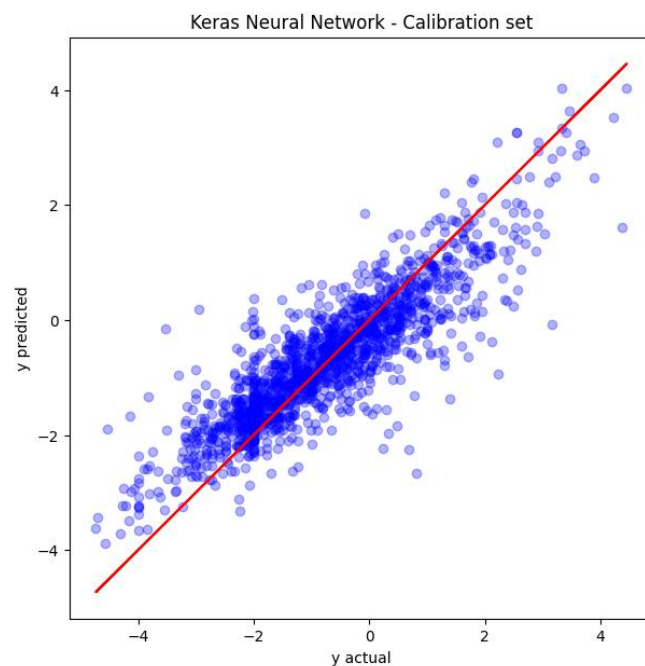


Figure S35. Graphical visualization of the results predicted by the KNN model *versus* the actual results for the calibration set. Keras neural network regression model was optimized with the Optuna package and with features to prevent overfitting; architecture: input layer with 70 neurons, 5 hidden layers, and 1 output layer (70-128-256-128-32-1); activation functions = relu, linear; dropout rate = 0.1, optimizer: Adam; learning rate = 0.001; validation split rate = 0.2; maximum training epochs = 20, batch size = 32, MSE = loss function; MAE = evaluation metric; early stopping feature.